

HARSHITH REDDY TADURI

tharshithreddy433@gmail.com | 9676210774 | Hyderabad, Telangana
<https://www.linkedin.com/in/harshithreddytaduri> | <https://github.com/Harshith-Reddy01>
<https://harshithreddy-portfolio.lovable.app>

CAREER SUMMARY

AI/ML and Data Science Engineer with hands-on experience in Machine Learning, Deep Learning, Generative AI and Data Analysis. Developed end-to-end AI and data-driven solutions across Cloud Infrastructure, Cybersecurity, Computer Vision and Healthcare. Skilled in Python, SQL, Deep Learning, Large Language Models (LLMs), Data Analysis, AI model development and evaluation. Interested in learning emerging technologies, developing scalable data-driven solutions for real-world problems.

SKILLS

- **Programming Languages** - Python, SQL, Bash
 - **Data Analysis & Visualization** - Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Power BI (DAX, Data Modeling, Interactive Dashboards, KPI Reporting)
 - **Machine Learning & Generative AI** - Scikit-learn, TensorFlow, Keras, Deep Learning (ANN, CNN, LSTM, GNN), Transformers, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Prompt Engineering, Feature Engineering, Model Training and Evaluation, Retrieval-Augmented Generation (RAG), Vector Databases
 - **Databases** - MySQL, SQL Server, PostgreSQL, Database Management Systems (DBMS)
 - **Tools & Technologies** - Git, GitHub, Jupyter Notebook, Hugging Face, Docker, SQLAlchemy, FastAPI, Terraform, Linux
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EDUCATION

- **Bachelor of Technology** - CSE (Artificial Intelligence and Machine Learning) | ACE Engineering College | 2021- 2025 | GPA: 7.83
 - **Intermediate** - Mathematics, Physics and Chemistry | Sri Chaitanya Junior College | 2019 - 2021. GPA: 9.66
 - **10th Standard** - SSC | Joshis Vidyaniketan School | 2018 - 2019. | GPA: 9.8
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EXPERIENCE

AI Intern | SYNYCS | June 2026 | Hyderabad, Telangana

- Designed, developed, and successfully delivered an end-to-end AI-powered Infrastructure-as-Code (IaC) Drift Detection platform from scratch as the key deliverable.
- Translated project requirements into a scalable AI solution by implementing backend services, cloud automation workflows, and intelligent infrastructure analysis.
- Integrated modern AI and cloud technologies to enable infrastructure governance, compliance validation, policy enforcement, and intelligent drift analysis.
- Successfully demonstrated and delivered the solution, gaining hands-on experience across the complete software development lifecycle, from system design to deployment.

Data Science Intern | Infoz IT Solutions | Mar 2026 - May 2026 | Hyderabad, Telangana

- Performed EDA and data preprocessing on real-world Kaggle datasets using Pandas, NumPy, Matplotlib, and Seaborn to identify patterns, handle outliers, and prepare data for machine learning.
 - Built and evaluated classification and regression models using Scikit-learn, applying feature engineering and hyperparameter tuning to improve model performance.
 - Developed interactive Power BI dashboards to visualize model results and communicate data-driven insights effectively to technical and non-technical stakeholders.
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PROJECTS

Autonomous Infrastructure as Code Generator and Drift Detector

Technologies: Python, FastAPI, LLMs (Ollama/Groq), PyTorch Geometric (GNN), PostgreSQL, SQLALchemy, Redis, Celery, Docker, Terraform, Keycloak, Next.js, Git

- **Designed, developed, and deployed an end-to-end AI-powered Infrastructure-as-Code (IaC) Drift Detection platform from scratch**, automating cloud infrastructure monitoring and governance.
- **Integrated LLMs** to enable natural-language IaC generation, intelligent drift analysis, and AI-driven remediation recommendations for cloud infrastructure.
- **Engineered a scalable backend** with FastAPI, PostgreSQL, Redis, Celery, and Docker by building REST APIs, asynchronous task pipelines, and a containerized multi-service architecture.
- **Leveraged Terraform, Open Policy Agent (OPA), and Graph Neural Networks (GNNs)** to automate compliance validation, policy enforcement, infrastructure impact (blast-radius) prediction, and cloud governance.

Hybrid Machine Learning Model for Efficient Botnet Detection

Technologies: Python, Deep Learning, CNN, RNN, LSTM, ANN

- Developed a **stacked deep learning model using CNN, RNN, LSTM, and ANN layers** to detect botnet attacks in IoT environments.
- Performed **data preprocessing, feature engineering, and model training** to capture complex botnet behavior patterns with 98% detection accuracy.
- Built a Flask-based web application with MySQL backend for real-time attack prediction, enabling security monitoring and scalable model interface.

Crowd Management Using Existing CCTV Network

Technologies: Python, YOLOv8, Deep Learning

- Built a **YOLOv8-based model** for real-time people detection and counting from CCTV.
- Implemented **real-time crowd monitoring** to track the number of individuals in public environments.
- Evaluated model performance using precision, recall, and real-time inference metrics to ensure reliable object detection.

CERTIFICATIONS

- **IIT GUWAHATI** - Data Science with Artificial Intelligence, Generative AI & Prompt Engineering, E&ICT Academy, IIT Guwahati
- **CISCO Networking Academy Certifications** - CLA, PCAP
- **Industry Exposure Program** - Infoz IT
- **Cambridge University** – Business English Certificate

LANGUAGES

- English
- Telugu
- Hindi